NATIONAL OSTEOPOROSIS FOUNDATION

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NOF Wants to Hear From You!

Tell us what you think about *How Strong Are Your Bones?* Take our online survey at http://nof.org/survey/howstrong. It will only take a few minutes to complete. Your feedback will help us to better meet your needs as we revise our existing publications and develop new ones.

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NATIONAL OSTEOPOROSIS FOUNDATION

How **strong** are your **bones?**

How Strong Are Your Bones? can help you understand a bone disease called osteoporosis, what increases your chance of getting it and how you can protect your bones. This brochure also explains bone mineral density (BMD) testing, which helps you find out if you have osteoporosis. It provides information about when to have a BMD test, what the results mean and when to consider treatment.

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Established in 1984, the National Osteoporosis Foundation (NOF) is the nation's leading voluntary health organization solely dedicated to osteoporosis and bone health.

National Osteoporosis Foundation's Vision
To make bone health a reality and a lifelong priority for all individuals.

National Osteoporosis Foundation's Mission
To prevent osteoporosis and related fractures, to promote lifelong bone health, to help improve the lives of those affected by osteoporosis and to find a cure through programs of awareness, advocacy, public and health professional education and research.

The National Osteoporosis Foundation (NOF) offers the following publications and services:

- **Free** brochures and handouts for people of all ages and backgrounds, including information in Spanish
- **Free** osteoporosis support groups
- **Free** online osteoporosis community
- Materials for community events and health fairs
- 100-page handbook, *Boning Up on Osteoporosis,* with up-to-date information
- Exercise video/DVD
- Online walking program and other special events
- Individual memberships that include a quarterly newsletter and much more!

To request information, become an NOF member or learn more about these programs and services, contact NOF at:

(800) 231-4222 (Toll-free)
(202) 223-2226 (Main)
(202) 223-2237 (Fax)
E-mail: request@nof.org
Web site: www.nof.org

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For More Information

Calcium-Rich Recipes. The Web sites below have recipes that will help you prepare meals high in calcium.

- http://recipefinder.nal.usda.gov/
- www.girlshealth.gov/bones/stayingstrong/calcium_cool.html
- http://ag.arizona.edu/maricopa/fcs/bb/recipes.html

Dietary Supplements. The Office of Dietary Supplements (ODS) of the National Institutes of Health offers information about dietary supplements in its "Dietary Supplement Fact Sheets." Here is the Web site and phone number:

- www.ods.od.nih.gov (Choose "Dietary Supplement Fact Sheets" under "Quick Links" on the side bar.)
- (301) 435-2920

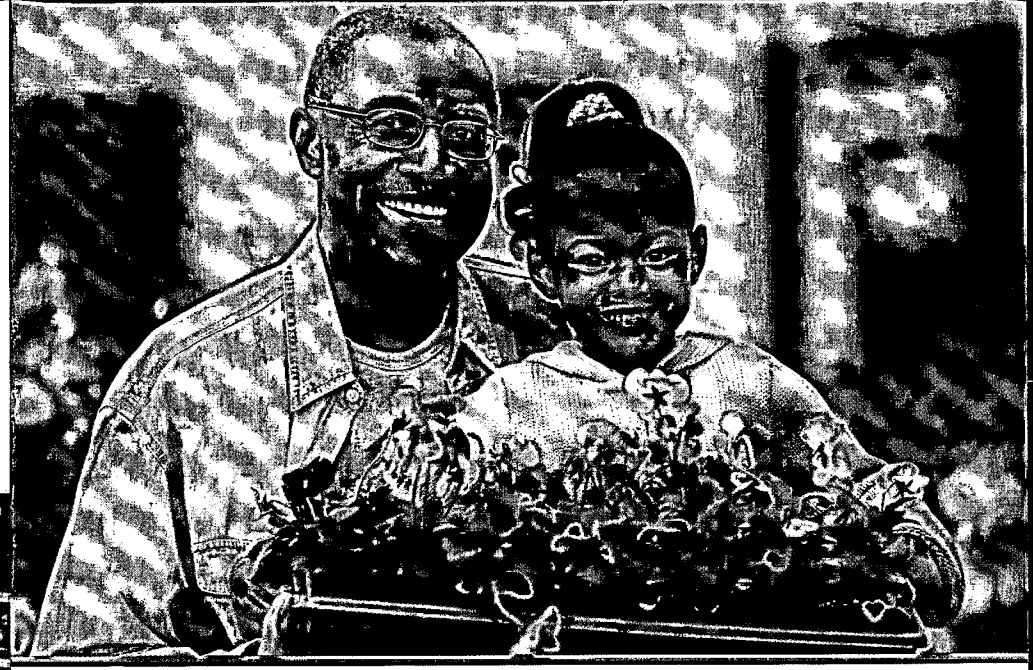
Medicines. For information on medicines, the following Web sites may be helpful:

- DrugDigest—www.drugdigest.org
- Food and Drug Administration (FDA)—www.fda.gov
- MedlinePlus—www.nlm.nih.gov/medlineplus/druginformation.html
- National Osteoporosis Foundation (NOF)—www.nof.org
- Physicians' Desk Reference (PDR)—www.pdrhealth.com

Physical Activity. The U.S. Department of Health and Human Services has *Physical Activity Guidelines for Americans*. It offers guidelines for children, adolescents, adults, older adults, women during pregnancy and people with disabilities. Here is the Web site and phone number:

- www.health.gov/paguidelines
- (800) 336-4797

Note: The Web sites and phone numbers listed are current at the time of publication.



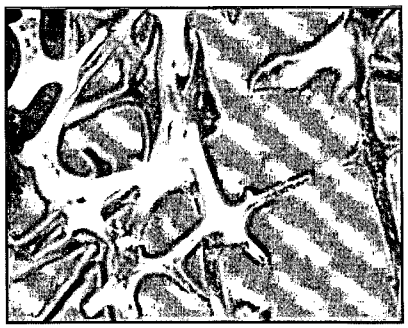
What Is Osteoporosis?

Osteoporosis means "porous bone." Bone under a microscope looks like a honeycomb. If you have osteoporosis, your bones have bigger holes and spaces in the honeycomb than healthy bone. This means your bones have lost density or mass. As a result, your bones become weak and break more easily.

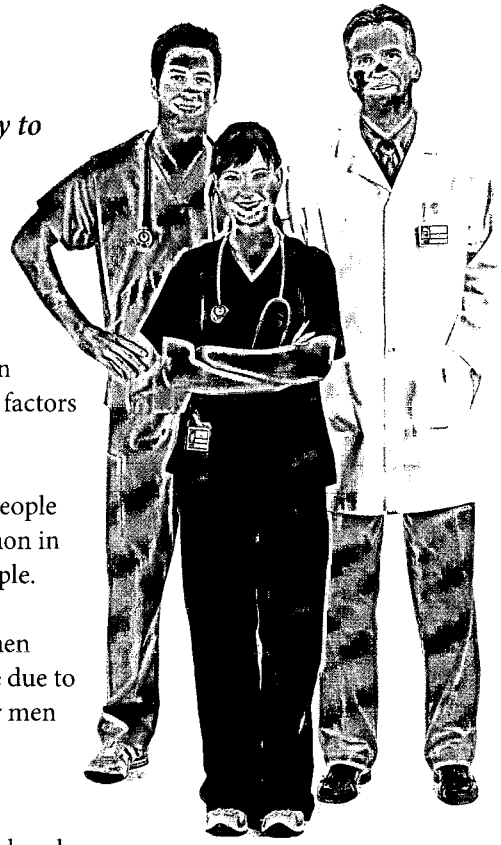
People with osteoporosis most often break a bone in the hip, spine or wrist. But, they can also break other bones. Sometimes bones break simply by lifting a child, bumping into furniture or sneezing.

Healthy bone

Bone with osteoporosis



Are You at Risk?



Some people are more likely to get osteoporosis. Things that increase your chance of having osteoporosis and the broken bones it can cause are called "risk factors." While you have no control over some risk factors, there are others you can change. Here are common risk factors for osteoporosis:

Age. Osteoporosis can affect people of all ages, but it is more common in older people than younger people.

Gender. About half of all women older than 50 will break a bone due to osteoporosis. Up to one in four men will too.

Family history. If one of your parents had osteoporosis or broke a bone as an adult, you are more likely to get osteoporosis.

Being small and thin. Women and men who are thin or have small bones are more likely to get osteoporosis.

Estrogen levels. Estrogen is a female hormone. When estrogen levels drop, many women lose bone density. This drop happens when a woman goes through menopause or has her ovaries removed. Teen girls and young women who often miss their periods usually have low estrogen levels. As a result, they may build less bone, and their bones are not as strong. Men also have small amounts of estrogen, which may help keep their bones healthy.

Testosterone levels. Testosterone is a male hormone. Low levels of testosterone in men can lead to osteoporosis.

To get your absolute fracture risk using the FRAX® tool, your healthcare provider enters information about your hip BMD and certain risk factors for osteoporosis. More healthcare providers will use the FRAX® tool when it becomes part of the software in DXA machines.

When is FRAX® used? Healthcare providers use the FRAX® tool to get the absolute fracture risk for a specific group of patients. They are:

- postmenopausal women or men age 50 and older
- people who have never taken an osteoporosis medicine
- people with low bone density (osteopenia)

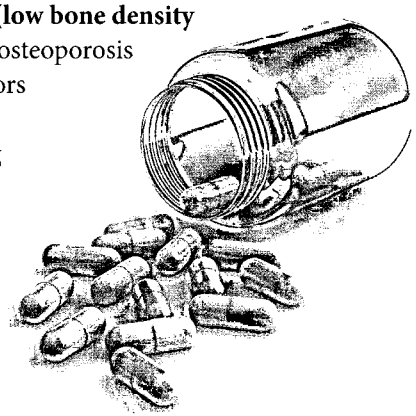
The FRAX® tool is not meant for adults under age 40 or children. It is also not necessary for people diagnosed with osteoporosis since it is already known that they are at greater risk for breaking a bone.

When to Take an Osteoporosis Medicine

The results of your BMD test will help your healthcare provider decide what you can do to prevent or treat osteoporosis. Your healthcare provider will also look at your risk factors for osteoporosis, your likelihood of breaking a bone, your medical history and your current health.

Treatment guidelines. Below are treatment guidelines for postmenopausal women and men age 50 or older:

- Most people with T-scores of -1 and above **(normal bone density)** do not need to take an osteoporosis medicine.
- People with T-scores between -1 and -2.5 **(low bone density or osteopenia)** should consider taking an osteoporosis medicine when they have certain risk factors including an increased chance of breaking a bone within the next 10 years (according to the FRAX® tool).
- All people with T-scores of -2.5 and below **(osteoporosis)** should consider taking an osteoporosis medicine.
- All people who have broken a hip or bones in the spine should consider taking an osteoporosis medicine.



Sometimes, a T-score can be normal in one area of bone and show osteoporosis in another area of bone or a different bone. The lowest T-score is used to make a diagnosis of osteoporosis. The lower a person's T-score, the lower the bone density. Here's what your T-score means:

- A T-score of -1 and above is **normal bone density**.
- A T-score between -1 and -2.5 is **low bone density** or **osteopenia**.
- A T-score of -2.5 and below is **osteoporosis**.

What your Z-score means. Your BMD test result also includes a Z-score that compares your bone density to what is normal in someone your age. Most experts recommend using only Z-scores for children, teens, women still having periods and younger men. A Z-score above -2.0 is normal according to the International Society for Clinical Densitometry (ISCD). A diagnosis of osteoporosis for younger age groups should not be based on a BMD test result alone.

Among older adults, low bone density is common, so Z-scores may be misleading. An older person may have a normal Z-score but still be at high risk for breaking a bone.

Absolute Fracture Risk

Healthcare providers know when to treat people who have osteoporosis with an osteoporosis medicine. But, it is less clear when to use these medicines in people with low bone density (osteopenia). For people with osteopenia, a method called absolute fracture risk provides a way to estimate the chance of breaking a bone within the next 10 years. This makes it easier to decide if a person should take an osteoporosis medicine. Calculating absolute fracture risk helps make sure that people with the greatest chance of breaking a bone are treated.

What is FRAX®? Healthcare providers can find out the absolute fracture risk of their patients with osteopenia by using a special Web-based tool on a computer in their office. This tool is called FRAX® (Fracture Risk Assessment Tool). It was developed to help healthcare providers figure out the chance that a patient will break a hip or other major bone within the next 10 years. The FRAX® tool is built into some central DXA machines. These DXA machines provide the same prediction as the Web-based tool.

Medicines. Some medicines can cause bone loss and osteoporosis. One type that is a major concern for bones is steroid medicines. Many people need to take them for rheumatoid arthritis, asthma or other conditions.

Ask your healthcare provider if any medicines you take can affect your bones. If you need to take a medicine that can cause bone loss, talk to your healthcare provider about taking the lowest possible dose for the shortest amount of time for your condition. Do not stop any treatment or change the dose of your medicine unless your healthcare provider says it's safe to do so.

Diseases and conditions. Many health problems can increase your chance of getting osteoporosis. Ask your healthcare provider if you have any diseases or conditions that can cause bone loss. Sometimes, treating a health problem that causes bone loss can improve your bone health.

List of Medicines That May Cause Bone Loss

- Antacids that have aluminum
- Antiseizure medicines (only some) such as Dilantin® or phenobarbital
- Aromatase inhibitors such as Arimidex®, Aromasin® and Femara®
- Cancer chemotherapeutic drugs
- Cyclosporine A and FK506 (tacrolimus)
- Gonadotropin releasing hormone (GnRH) such as Lupron® and Zoladex®
- Heparin
- Lithium
- Medroxyprogesterone acetate (Depo-Provera®)
- Methotrexate
- Proton pump inhibitors (PPIs) such as Nexium®, Prevacid® and Prilosec®
- Selective serotonin reuptake inhibitors (SSRIs) such as Lexapro®, Prozac® and Zoloft®
- Steroid medicines such as cortisone and prednisone
- Tamoxifen (premenopausal use)
- Thiazolidinediones (Actos® and Avandia®)
- Thyroid hormones in excess

Note: This list may not include all medicines that may cause bone loss.

List of Diseases and Conditions That May Cause Bone Loss

AIDS/HIV

Ankylosing spondylitis

Blood and bone marrow disorders

Breast cancer

Chronic obstructive pulmonary disease (COPD), including emphysema

Cushing's syndrome

Depression

Diabetes

Eating disorders, especially anorexia nervosa

Female athlete triad (includes loss of menstrual periods, an eating disorder and excessive exercise)

Gastrectomy

Gastrointestinal bypass procedures

Hyperparathyroidism

Hyperthyroidism

Inflammatory bowel disease, including Crohn's disease and ulcerative colitis

Kidney disease that is chronic and long lasting

Liver disease that is severe, including biliary cirrhosis

Lupus

Lymphoma and leukemia

Malabsorption syndromes, including celiac disease

Multiple myeloma

Multiple sclerosis

Organ transplants

Parkinson's disease

Polio and post-polio syndrome

Poor diet, including malnutrition

Premature menopause

Prostate cancer

Rheumatoid arthritis

Scoliosis

Spinal cord injuries

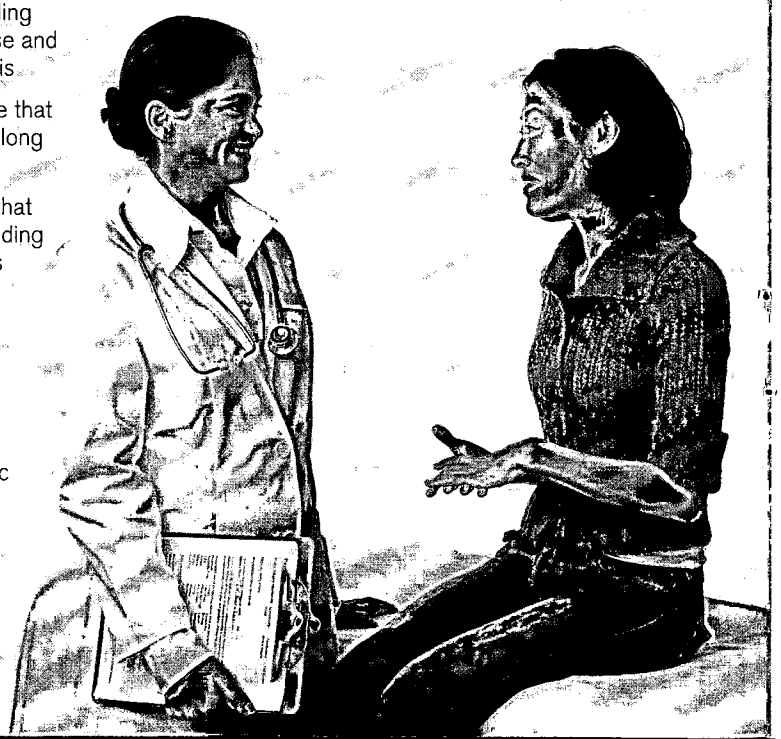
Stroke

Thalassemia

Thyrotoxicosis

Weight loss

Note: This list may not include all diseases and conditions that may cause bone loss.



Once you break a bone in the spine, you are more likely to break another bone in the spine. Talk with your healthcare provider about steps to protect your spine. You should also think about taking an osteoporosis medicine. When you break a bone in the spine, you still need to have a BMD test if you haven't had one.

Understanding Your BMD Test Results

When you have a BMD test, your results will show a T-score and a Z-score.

What your T-score means. Healthcare providers use the T-score to diagnose osteoporosis in women going through menopause, postmenopausal women and men age 50 and older. It shows how much your bone density is above or below normal compared to an average 30-year-old adult who is healthy. T-scores are not used in children and teens because their bones are still growing.

Guide to T-scores

Category	T-scores	
	Range	Examples
Normal Bone Density	-1 and above	+1.0 +0.5 0 -0.5 -1.0
Low Bone Density (Osteopenia)	Between -1 and -2.5	-1.1 -1.5 -1.8 -2.0 -2.4
Osteoporosis	-2.5 and below	-2.5 -2.8 -3.0 -3.5 -4.0

Note: An increase or decrease of 1.0 in a T-score equals a 10–15 percent change in bone density. For example, a person with a T-score of -2.5 has 10–15 percent less bone density than a person with a T-score of -1.5.

When repeating a BMD test, it's best to use the same type of DXA machine and have it done at the same place. This provides the best comparison with your last test result. Although it is not always possible to do this, it's still important to compare your current BMD test to your last one.

Paying for your BMD test. Medicare pays for a BMD test every two years and sometimes more often. Many insurance plans pay for BMD tests, depending on a person's age and reasons for having it done. Your insurance plan should be able to tell you if it will pay for a BMD test.

Other Tests You May Need

Lab tests. If you are diagnosed with osteoporosis, your healthcare provider may do one or more lab tests to find out what is happening in your body. There are different lab tests your healthcare provider may do. Some of them can help you:

- Know if you are getting enough vitamin D
- Find out if you have a condition that can cause bone loss
- Learn if you're losing bone faster than normal
- See if your osteoporosis medicine is working

Keep in mind that lab tests don't take the place of BMD testing.

Tests to find broken bones. An x-ray is the most common way to tell if you have a broken bone. Sometimes there is no pain with broken bones in the spine. Your healthcare provider may order an x-ray of your spine if you: (1) lose one-half inch or more in height in one year, (2) have stooped or hunched posture, (3) develop back pain or (4) have lost more than 1½ inches from your original height.

Another test to find broken bones in the spine is a vertebral fracture assessment (VFA). A VFA is done with a DXA machine and uses less radiation than a standard x-ray. VFAs can show these breaks in the spine.

Protecting Your Bones

You can take steps at any age to protect your bones. Making healthy lifestyle choices can help prevent osteoporosis and the painful broken bones it can cause. Here's what you can do to keep your bones strong and healthy:

Calcium. Getting enough calcium helps build strong bones when you are younger and keeps them strong later in life. Dairy foods are high in calcium, while some green vegetables and other foods have smaller amounts of calcium. If you get the recommended amount of calcium, you don't need to take a calcium supplement. If you need a calcium supplement, your doctor, pharmacist or other healthcare provider can help you choose one that is right for you.

Vitamin D. Your body needs vitamin D to absorb calcium. You can get small amounts of vitamin D from a few foods. These include egg yolks, liver, fortified milk and fatty fish such as mackerel, salmon and tuna. You can also get vitamin D from the sun, but you need to protect your skin from too much sun. Using sunscreen or sunblock reduces the amount of vitamin D you get from the sun. So, many people need to take a supplement. If you're not sure you're getting enough vitamin D, talk to your doctor or other healthcare provider. Ask if you should have a simple blood test to check your vitamin D level.

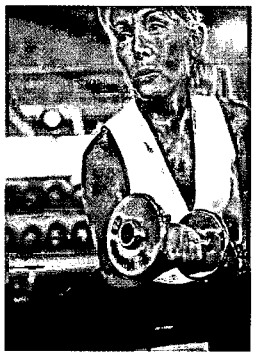
NOF's Calcium and Vitamin D Recommendations

Adults under age 50
need 1,000 mg of calcium and 400–800 IU of vitamin D every day.

Adults age 50 and older
need 1,200 mg of calcium and 800–1,000 IU of vitamin D every day.

Other vitamins, minerals and nutrients. Eating a well-balanced diet that includes fruits and vegetables is good for your bones. Very high amounts of protein, salt (sodium) or caffeine can cause bone loss. You can help prevent some of this loss by getting the amount of calcium your body needs. Keep in mind that it's still important to get enough protein for your bones and overall health.

Exercise. Your bones get stronger when you make them work. People who are bedridden or do not exercise are at high risk for osteoporosis. You should exercise at least 2½ hours every week. That's 150 minutes, and more



is even better. Try to make exercise part of your routine every day. Two types of exercises help keep your bones strong and healthy:

Weight-bearing exercises. Examples are dancing, hiking, jogging, elliptical training machines, aerobics and walking. Although swimming and biking are good for overall health, they are not weight-bearing exercises.

Muscle-strengthening exercises. Examples are lifting weights, using elastic exercise bands, lifting your own body weight or using weight machines. Pilates and yoga are also muscle-strengthening exercises.

Check with your doctor or other healthcare provider before beginning a new exercise program to make sure it's right for you. Your healthcare provider may also recommend you see a physical therapist to teach you exercises that are safe and healthy for your bones.

Smoking and alcohol. Don't smoke or drink too much alcohol. Smoking and drinking three or more alcoholic drinks a day are bad for your bones.

Falls. Many people break a bone after they fall. So preventing falls is important. Muscle-strengthening exercises improve balance and coordination and help prevent falls. Balance training exercises and activities such as Tai Chi can also help prevent falls. Some medicines can increase your chance of falling, especially if they make you sleepy or drowsy. Some diseases can also increase your chance of falling. You can make your home safe from falls by removing clutter, placing grab bars in the bathroom and having plenty of light. Wear shoes with rubber soles and low heels. Be careful both indoors and outdoors.

Yearly check up. Make an appointment with your doctor or other healthcare provider at least once a year. If your healthcare provider hasn't talked to you about your bone health yet, it's time for you to bring it up. Make your healthcare provider your partner in keeping your bones strong and healthy.

When you go for your appointment, be sure to take the prescription with you to give to the testing facility. The testing facility will send your BMD test results to your healthcare provider, who will then share the results with you.

How often to have a BMD test. NOF recommends that people taking an osteoporosis medicine repeat their BMD test by central DXA every two years. Some healthcare providers may have certain patients repeat their BMD test after one year. Peripheral tests (pDXA, QUS and pQCT) should not be used to see if an osteoporosis medicine is working.

If you have a normal BMD test result, your healthcare provider will recommend when to have another one based on your age and risk factors for osteoporosis.

People of Larger Size

BMD test. Most central DXA machines cannot measure bone density in the hip and spine of people who weigh over 300 pounds. Some newer machines can measure bone density in people who weigh up to 400 pounds, but they are not widely available. For people of larger size, some healthcare providers recommend getting both a peripheral DXA test of the radius bone of the forearm (between the wrist and elbow) and a peripheral test of the heel or another bone. Doing both of these tests might give more complete information.

Weight loss. In general, heavier people have a lower risk for osteoporosis. However, being larger does not always mean a person is protected from osteoporosis. Because obesity is linked to heart disease, diabetes and other health problems, achieving a normal weight has many health benefits. But, weight loss can increase the risk of bone loss. So it's important to get enough calcium, vitamin D and exercise while dieting. This will help people protect their bones while losing weight.

When starting a weight loss program, healthcare providers may do a BMD test, especially if a person is thinking about a surgical weight loss procedure. These test results will be compared with future BMD tests to check for changes in bone density.

Peripheral tests are helpful for predicting the chance of breaking a bone when a DXA test is not available. But, they are not used to diagnose osteoporosis. These tests are often done at health fairs and pharmacies as well as some medical offices. The results of peripheral tests and central DXA tests cannot be compared.

Are you a postmenopausal woman or man age 50 and older? Have you recently broken a bone? If you answered "yes" to both questions, you should talk to your doctor or other healthcare provider about getting a BMD test if you've never had one.

If you have abnormal results from a peripheral BMD screening, you should see your healthcare provider. Discuss whether you need a central DXA test of the hip and/or spine.

Safety of BMD tests. A BMD test is safe and painless. With most types of BMD tests, you are fully dressed. The test usually takes less than 15 minutes. BMD tests are non-invasive. This means that no needles or instruments are placed through the skin or in the body.

The QUS method of BMD testing uses sound waves to measure bone density. The DXA machine and most other types of equipment use radiation. The amount of radiation used with a DXA machine is very small. In fact, flying in a plane round trip between New York and San Francisco exposes you to 10-15 times more radiation. QCT and pQCT use more radiation than DXA and other BMD tests, but are still safe.

Conditions affecting your BMD test. There are certain conditions that may cause your bone density to seem better than it is. These include osteoarthritis of the spine, scoliosis and broken bones in the spine. Certain types of surgery of your spine may also affect your BMD test results. Talk with your healthcare provider about your BMD test results if you have any of these conditions.

Where to have a BMD test. In most states, you will need to get a prescription (also called an order or referral) from your healthcare provider to have a BMD test done. Your healthcare provider may suggest a place to go for your BMD test. Most hospital radiology departments, private radiology groups and some medical practices offer BMD testing. Your health insurance plan may also be able to suggest places to have a BMD test.

Diagnosing Osteoporosis

Learning if you have osteoporosis may involve several steps during your visit with your healthcare provider. These include your medical history, a physical exam and a bone mineral density (BMD) test.

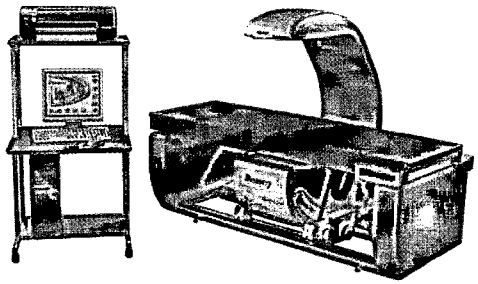
Medical history. Be prepared to answer questions about your health, medicines you take and the health of your family members. You may also be asked about your diet, exercise habits and if you smoke or drink alcohol. This information helps your healthcare provider learn if you have risk factors for osteoporosis.

Physical exam. Once you are age 40 or older, your healthcare provider may measure you every year to see if you are losing height. Some healthcare providers also check the spine to see if it is curving forward. Losing height and changes in the spine can mean you have already broken several bones in the spine. Broken bones in the spine are often due to osteoporosis.

Bone mineral density (BMD) test. If you are a postmenopausal woman or man age 50 or older, be sure to ask your healthcare provider when you should have a BMD test. Depending on your age, medical history and physical exam, your healthcare provider will decide if you need a BMD test. It's how you find out if you have osteoporosis. A BMD test also lets you know if you have low bone density, which is also called osteopenia. Having low bone density means you're more likely to get osteoporosis. The next section provides more information about BMD testing.

Bone Mineral Density (BMD) Testing

A BMD test uses a special machine to measure bone density. This test lets you know the amount of bone you have in your hip, spine or other bones. When bones lose density, they are not as strong and can break more easily.



Most experts recommend a BMD test of the hip and spine for several reasons. First, people with osteoporosis have a greater chance of breaking these bones. Second, breaking these bones can cause more serious problems, including longer recovery time, greater pain and even disability. Third, the results of a BMD test in the hip and spine can predict your chance of breaking these and other bones in the future.

What a BMD test can do. A BMD test can help you and your healthcare provider:

- Learn if you have weak bones or osteoporosis before you break a bone
- Predict your chance of breaking a bone in the future
- Let you know if you have osteoporosis after you break a bone
- See if your bone density is improving, getting worse or remaining the same over time
- Decide if you need a medicine to help prevent broken bones
- Find out how well an osteoporosis medicine is working

Who should have a BMD test? NOF recommends a BMD test for these people:

- Women age 65 or older
- Postmenopausal women under age 65 with risk factors for osteoporosis (for risk factors, see page 2)
- Men age 70 or older
- Men age 50-69 with risk factors for osteoporosis
- Women going through menopause with certain risk factors
- Adults who break a bone after age 50
- Adults with certain medical conditions (for a list of diseases and conditions that may cause bone loss, see page 4)
- Adults taking certain medicines (for a list of medicines that may cause bone loss, see page 3)
- Postmenopausal women who have stopped taking estrogen therapy (ET) or hormone therapy (HT)

BMD testing in younger people. NOF does not recommend routine BMD testing for children, premenopausal women or men under age 50. BMD tests are usually only done for people in these age groups if they break several bones easily or break bones that are unusual for their age. For example, breaking a hip or a bone in the spine as a young person is unusual. Healthcare providers may recommend a BMD test for young adults who need to take a steroid medicine for three or more months.

Types of BMD tests. Standard x-rays are not used for BMD testing because they don't show osteoporosis until 25 to 40 percent of bone density has been lost. BMD tests can find the disease earlier. The types of BMD tests include:

Central DXA. The National Osteoporosis Foundation (NOF) recommends a BMD test of the hip and spine using a central DXA machine to diagnose osteoporosis. DXA stands for dual energy x-ray absorptiometry. When testing can't be done on the hip and spine, NOF suggests a peripheral DXA test of the radius. The radius bone is located between the wrist and elbow.

QCT. QCT stands for quantitative computed tomography. It measures bone density in both men and women. It can predict a woman's chance of breaking a bone in the spine. This test is sometimes done when a DXA machine is not available, but it is more expensive and not as good at measuring changes in bone density over time. QCT results cannot be compared to DXA results.

Screening tests. Screening tests measure bone density in the lower arm, wrist, finger or heel. They are also called peripheral tests. The types of peripheral BMD tests include:

- pDXA (peripheral dual energy x-ray absorptiometry)
- QUS (quantitative ultrasound)
- pQCT (peripheral quantitative computed tomography)